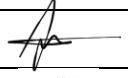


Assignment/Report Cover Sheet

Student Name	Student ID Number	Group No	Signature
ABDUL RAHIM BIN IDRIS	102368	GROUP 7	
PUTRA NURIMAN AKMAL BIN ROZIMAN	102917		
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Subject Code : TMF2954	Subject Name : Java Programming
Assignment Title : Project Design (SDG 4: Quality Education)	Lecturer : Madam Azima Alya
Due Date : 5 th June 2026	Date Submitted : 5 th June 2026

This cover sheet must be completed, signed and firmly attached to the front of the submission. All work must be submitted by the due date. If an extension of work is granted, an assignment extension acknowledgement slip must be signed by the lecturer/tutor and attached to assignment. Please note that it is your responsibility to retain copies of your assignment.

c

Plagiarism and Collusion are methods of cheating that falls under Peraturan Akademik Universiti Malaysia Sarawak para 11: Etika Akademik

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Plagiarism is the presentation of work which has been copied in whole or in part from another person's work, or from any other source such as the internet, published books or periodicals without due acknowledgement given in the text.

Collusion

Collusion is the presentation of work that is the result in whole or in part of unauthorized collaboration with another person or persons.

Where there are reasonable grounds for believing that cheating has occurred, the only action that may be taken when plagiarism or collusion is detected is for the staff member not to mark the item of work and to report or refer the matter to the Dean. This may result in work being disallowed and given a fail grade or if the circumstances warrant, the matter may be referred to a Committee of inquiry for investigation. Such investigation may result in the matter being referred to the University Discipline Committee, **which** has the power to exclude a student.

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1.0 UML Diagram

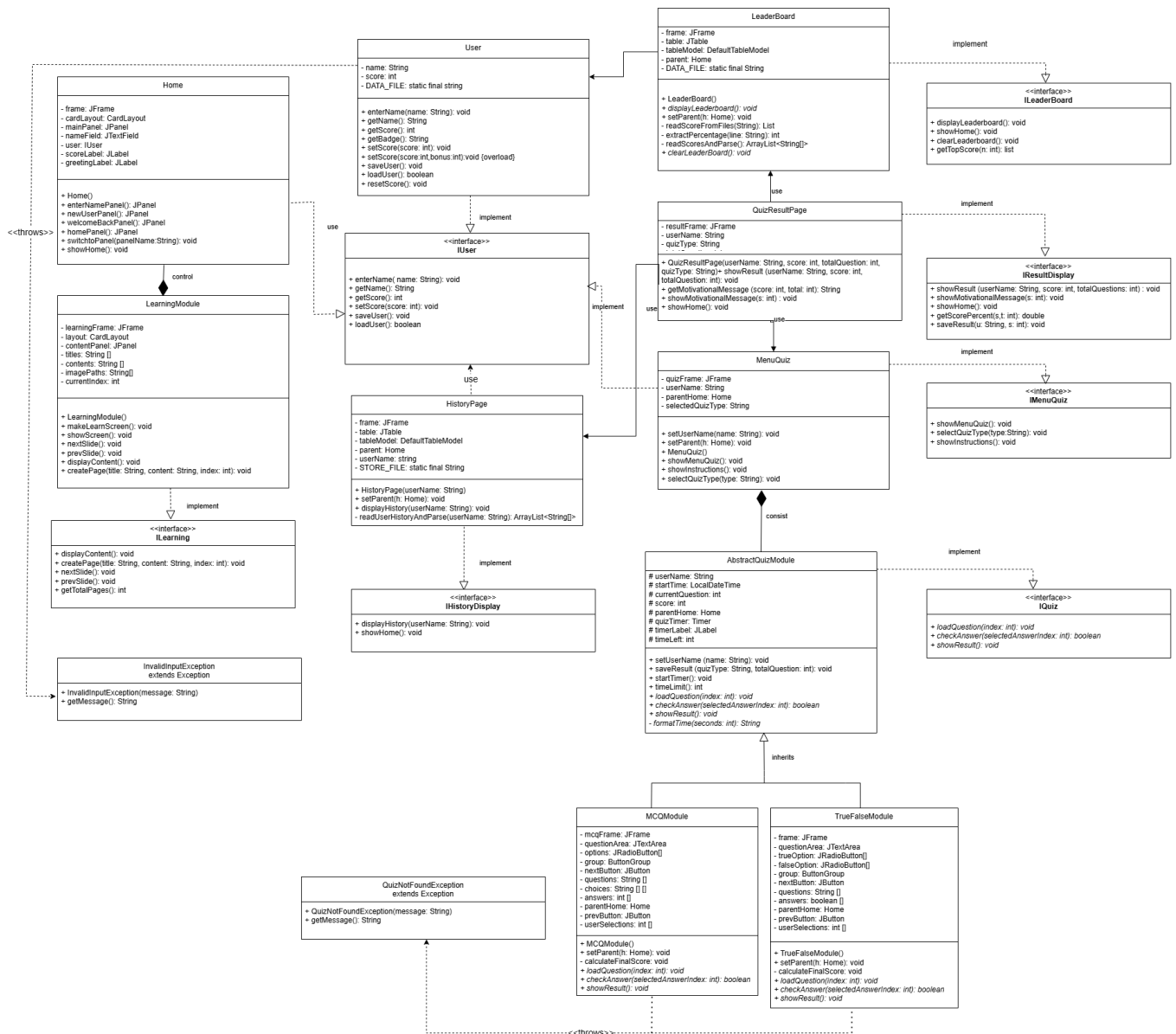


Figure 1.0 UML Diagram

Based on the UML Diagram above, there are 12 classes and 7 interfaces inside it. The 12 classes are: Home, User, LearningModule, MenuQuiz, AbstractQuizModule, MCQModule, TrueFalseModule, QuizResultPage, LeaderBoard, HistoryPage, InvalidInputException, and QuizNotFoundException. The 7 interfaces are: IUser, ILearning, IMenuQuiz, IQuiz, IResultDisplay, ILeaderBoard, and IHistoryDisplay. Besides, each class have more than 3 attributes and 3 methods. For example, classes like User contain 3 attributes and 8 methods. Every major component implements a dedicated interface, for example, User implements IUser, LeaderBoard implements ILeaderBoard.

Object-oriented Principles used in this project is Polymorphism, Inheritance and Abstraction.

- **Inheritance:** The AbstractQuizModule acts as a parent class. The MCQModule and TrueFalseModule are inherited from it. This establishes a clear, correct parent-child relationship and efficiently reuses the timer and scoring logic.

- Abstraction: The use of AbstractQuizModule ensures that common variables (like score and startTime) and methods (like startTimer()) are centralized, while forcing child classes to implement their own specific behaviors. The use of interfaces (IUser, ILearning, IQuiz, etc.) shows a strong grasp of abstracting behaviors from implementations.
- Polymorphism (Overloading): The User class contains method overloading. It features setScore(score: int) and an overloaded version setScore(score: int, bonus: int), to update scores with or without bonus points flexibly.
- Polymorphism (Overriding): The MCQModule and TrueFalseModule provide their own specific implementations of methods like checkAnswer() and showResult(), overriding the generalized behaviors to suit multiple-choice or true/false logic.

The diagram also includes CustomException which includes InvalidInputException (thrown by Home) and QuizNotFoundException (thrown by MenuQuiz). This ensures the program handles user input errors or missing files without crashing.

2.0 Classes and Interfaces Distribution

Name	Classes Implemented	Interfaces Implemented	Roles
Abdul Rahim Bin Idris	AbstractQuizModule, MCQModule, TrueFalseModule, QuizNotFoundException	IQuiz	Developing the quiz module and implementation of quiz types. Each quiz type consists of 30 questions. Include error handling if a quiz cannot be found.
Wan Adam Danish Bin Wan Abit	AbstractQuizModule, LearningModule, MenuQuiz	ILearning, IMenuQuiz	Assisting with the quiz module and creating learning module and menu quiz with its own interface.
Putra Nuriman Akmal Bin Roziman	LeaderBoard, HistoryPage	ILeaderBoard, IHistoryDisplay	Developing leaderboard and history tracking features. Handles score records, leaderboard rankings, user quiz history, and testing/debugging of the tracking module.
Qhairunnisa Shahirah Binti Zakaria	Home, User, QuizResultPage	IUser, IResultDisplay	In charge of main UI navigation & presentation. The window management, central navigation layout, menus, and result summaries.

Table 2.0 *Classes and Interfaces Task Distribution*

The table above shows the roles and responsibilities of each member. Each member handled the implementation of both the interfaces and classes which helped to construct the SDG 4: Quality Education Application. The application is built entirely around SDG 4: Quality Education, aiming to educate users on the importance of inclusive and equitable education and promote lifelong learning opportunities.

3.0 Proof of Discussions

